

EE 557 : Adaptive Control

Course Code : EE 557

Course Name : Adaptive Control

L-T-P-C : 3-0-0-3

Intended for : Engineering graduates (pursuing M. Tech, PhD)

Prerequisite : Basic (Level 1) course in Automatic Control, e.g., EE301 – Control Systems Engineering. Familiarity to some extent with Standard statespace description of dynamical systems and Digital Control is desirable.

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Introduction:** Feedback control system design steps, effect of process variations, adaptive schemes and applications. Models of dynamic systems (recapitulation only): State space and Input-output models, plant parametric models. (3 Hours)
- **Stability:** Norms, Quadratic forms, Positive definiteness of matrices; BIBO stability, Stability in the sense of Lyapunov, Stability of linear systems. (4 Hours)
- **Parameter Estimation:** Least squares and Regression models, Recursive estimation. (6 Hours)
- **Deterministic and Stochastic Self-tuning Regulators:** Pole placement design, Continuous-time self-tuners, Direct self-tuning regulators; Minimum-variance and moving average controllers, stochastic self-tuning regulators, Linear quadratic self-tuning regulator, Adaptive predictive control. (6 Hours)
- **Model Reference Adaptive Control:** Simple direct model reference adaptive control (MRAC) schemes, MRAC for SISO plants, Direct MRAC with unnormalized and normalized adaptive laws, Indirect MRAC with unnormalized and normalized adaptive laws, Adaptive laws with projection. (8 Hours)
- **Auto Tuning and Gain Scheduling:** Auto tuning of PID controllers, Design of gain scheduling controllers. (4 Hours)
- **Robust Adaptive Laws:** Plant uncertainties and robust control, Instability phenomena in Adaptive systems, Robust adaptive laws. (5 Hours)
- **Robust Adaptive Control Schemes:** Robust identifiers and adaptive observers, Robust MRAC, Robust adaptive pole placement control schemes. (4 Hours)
- **Perspectives on Adaptive Control:** Expert control systems, Learning systems, Future trends, Conclusion. (2 Hours)

Laboratory/practical/tutorial Modules:

- Not envisaged

Textbooks:

1. K. J. Astrom and B. Wittenmark, **Adaptive Control**, 2nd Edition, Addison Wesley, 1995 (2e), Dover Publications, 2008 (2e rev.)

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2. Petros A. Ioannou and Jing Sun, **Robust Adaptive Control**, Dover Publications, 2012
 3. S. Sastry and M. Bodson, **Adaptive Control: Stability, Convergence and Robustness**, Dover Publications, 2011
 4. R. Isermann and M. Munchhof, **Identification of Dynamic Systems: An introduction with applications**, Springer 2011

References:

1. NA